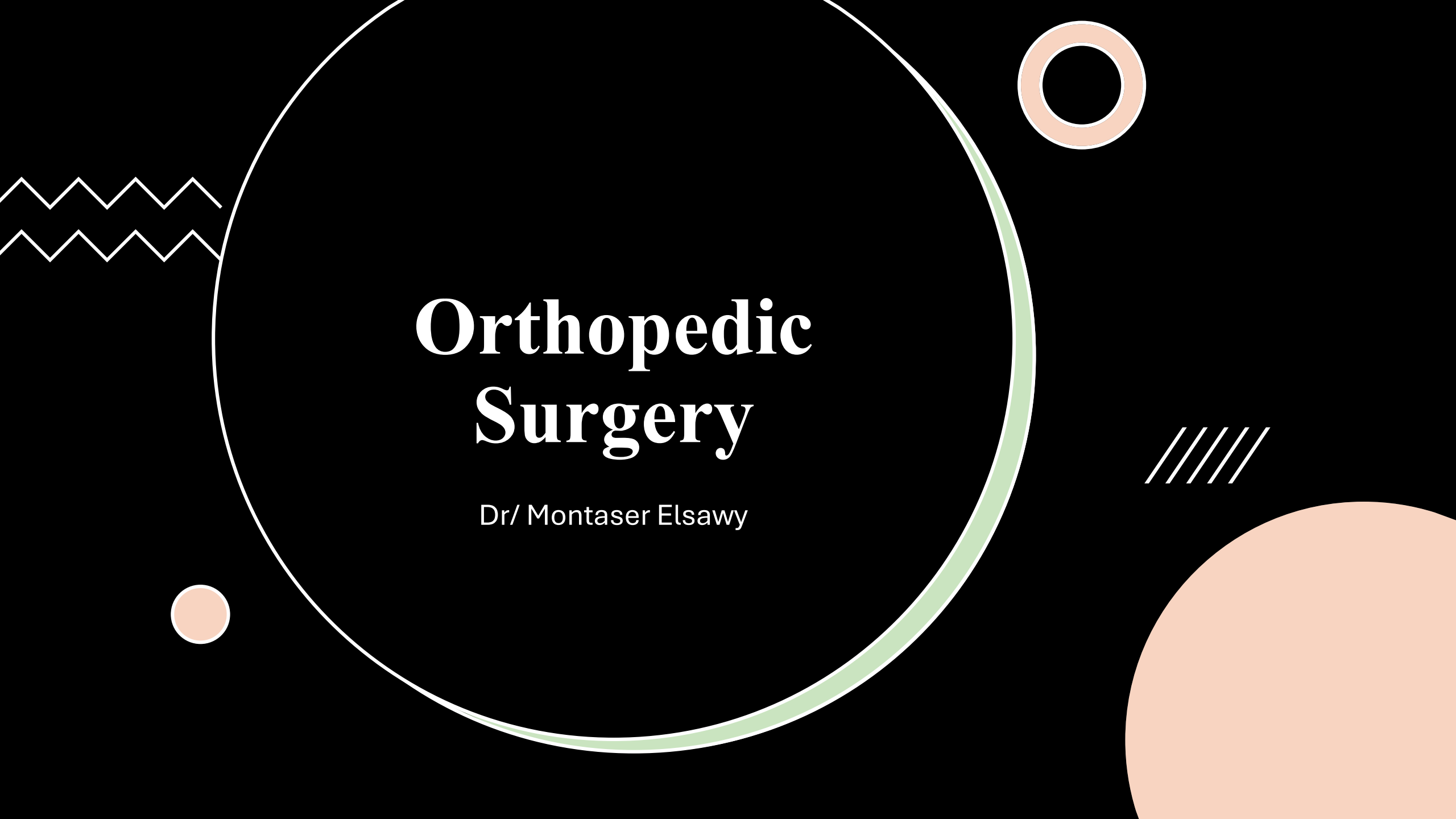




Orthopedic Surgery

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IMPORTANT DEFINITIONS

- **Fracture**: *discontinuity of bone.*
- **Dislocation**: disruption of the continuity of a joint. □ joint surface are no longer in continuity”
- **Subluxation**: partial disruption of the continuity of a joint. □ Joint surface are still opposed.”
- **Fracture dislocation**: dislocation together with a fracture of one or more of the bones forming the joint.

Joint dislocation

Reduction of the joint must be performed as soon as possible, either in the accident and emergency department under sedation (e.g. anterior dislocation of the shoulder), or in the operating theatre under general anaesthesia. In some cases where closed reduction fails, open surgical reduction is performed.



A fracture dislocation of the humeral head, showing the humeral shaft and head as separate pieces, as well as portions of the greater tuberosity. All are separate from the normal anatomical position of the femoral head, and the glenoid can be clearly seen in the background of the X-ray.

Initial Assessment of bone fractures

Advanced trauma life support guidelines	
Primary survey	
A	Airway with C-spine control
B	Breathing and ventilation
C	Circulation and haemorrhage control
D	Disability (neurological evaluation)
E	Exposure and environment
Secondary survey	
Once the resuscitation efforts are well established and the vital signs are normalising, the secondary survey can begin. This involves a complete history and physical examination, including the reassessment of all vital signs.	

Examination

Careful examination of each injured limb includes:

- *Skin.* Ascertain whether any skin breaks communicate with any underlying fracture.
- *Circulation.* The most common cause of absent pulses is kinking or compression of the artery by the fracture. Often, a reduction of the fracture (realigning the fracture into the anatomical position) or application of traction results in the return of perfusion.
- *Nerves.* Certain injuries may have a high associated risk of neurological injury, manifest by loss of power and/or sensation. For example, the axillary nerve is at risk from shoulder fracture dislocation and its integrity should be documented prior to reduction.
- *Joint above and below.* Often fractures can involve the joint immediately above or below. A complete examination of these joints is required as it may influence definitive treatment of the injury.

INVESTIGATIONS

Plain X-rays

Two views, in orthogonal planes including the joint above and below are generally used to evaluate almost all bone pathology and are performed weight-bearing whenever possible. Alignment (the degree of varus or valgus deformity), as well as true bone length, can be quantified. X-rays are also routinely used to confirm the correct position of bones, joints or prostheses after surgery.

Ultrasound

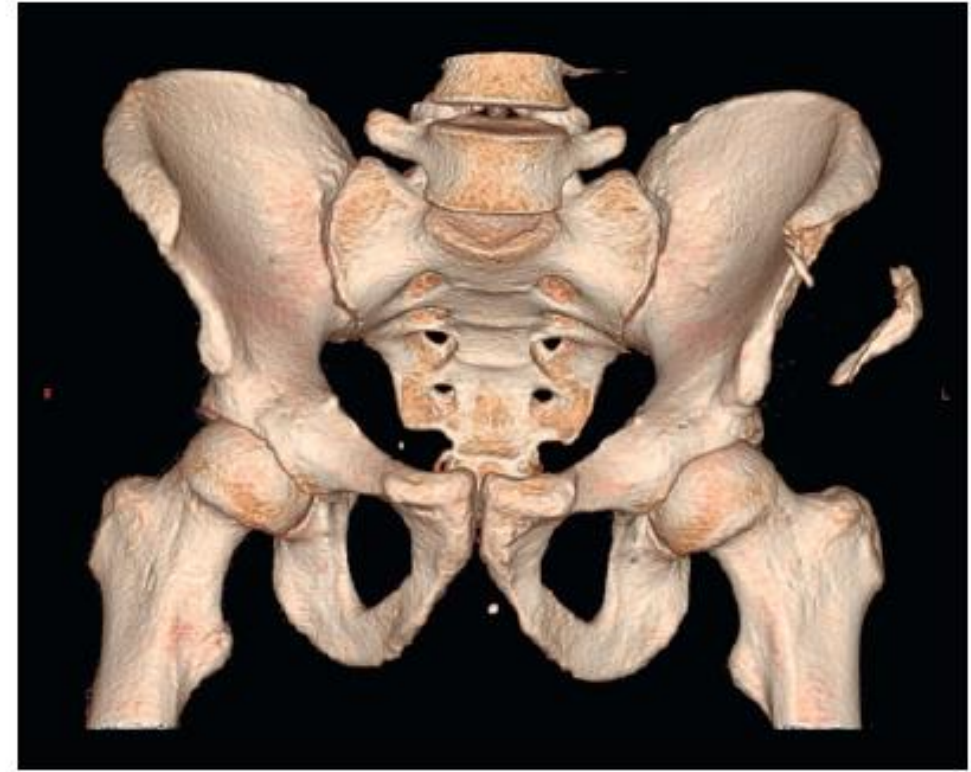
This is used frequently to evaluate soft tissue pathology (e.g. tendon ruptures, bleeds into soft tissues, other muscular or tendinous pathology) and to guide biopsy or injection. As a dynamic technique, it can be used to visualize the movements of tendons and muscles under direct vision.

Nerve conduction tests and electromyography (EMG)

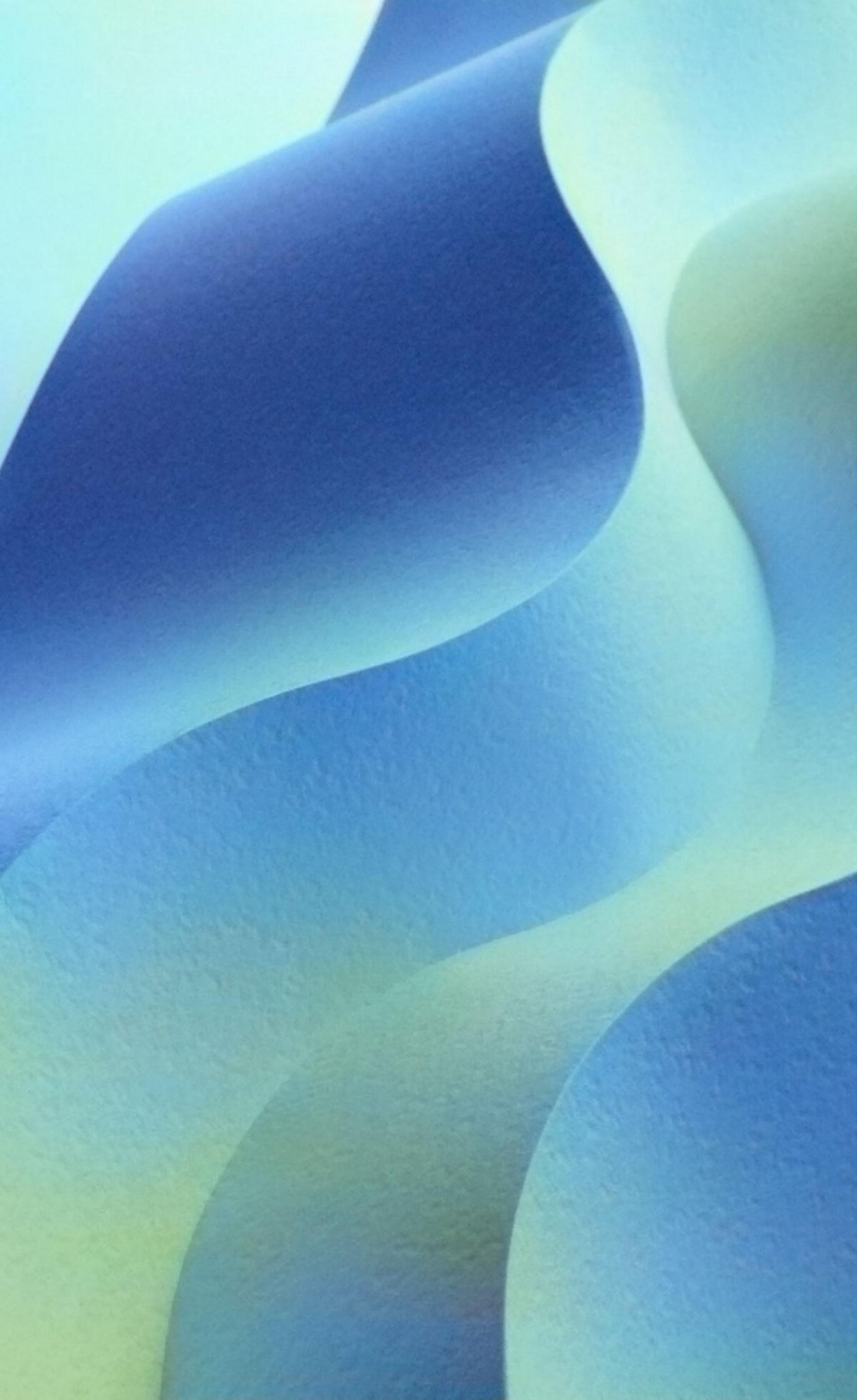
These are used to evaluate nerve entrapment syndromes, nerve injuries, neuropathies and abnormalities of muscular contraction.

Computed tomography (CT)

CT provides excellent images of bone anatomy and can be used to supply three-dimensional images to help in the reconstruction of complex fractures. The thin axial slices are particularly useful as a guide for obtaining biopsy specimens



A three dimensional CT showing a iliac wing fracture.



Fracture management

Classification

Fractures are usually classified by:

- whether they are in communication with the skin surface (open or compound) or not (closed).
- Their appearance on X-ray: for example:
- **A) Comminuted** (in multiple pieces, Fig.1, spiral (where the fracture curves in a large spiral around the long axis of the bone)
- **B) Transverse** (straight across the bone, Fig.2) This appearance usually correlates with the mechanism of injury – a twisting injury tends to cause a spiral fracture for example.
- **C) Anatomical site:** for example, ***intra-articular*** (involving the joint surface, Fig. 3), metaphyseal, epiphyseal or diaphyseal

Classification of Fractures



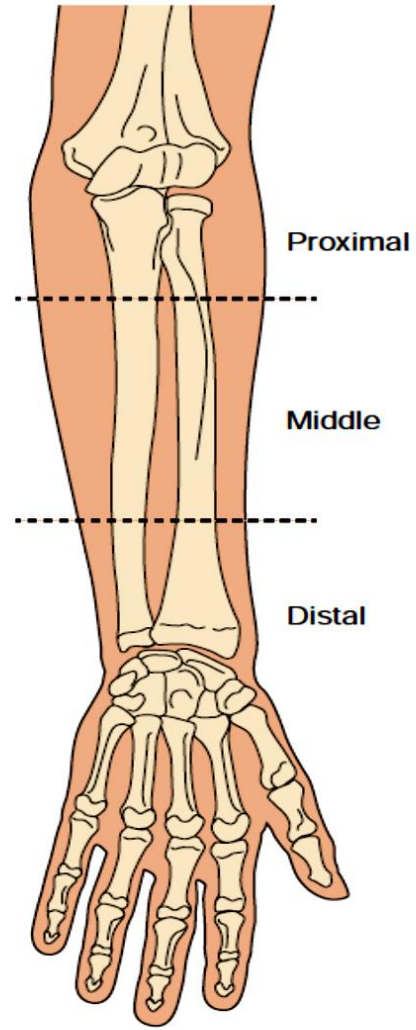
Fig.1 A comminuted fracture of the distal femur involving the knee joint as well. The femur can be seen to have broken into multiple (comminuted) fragments.



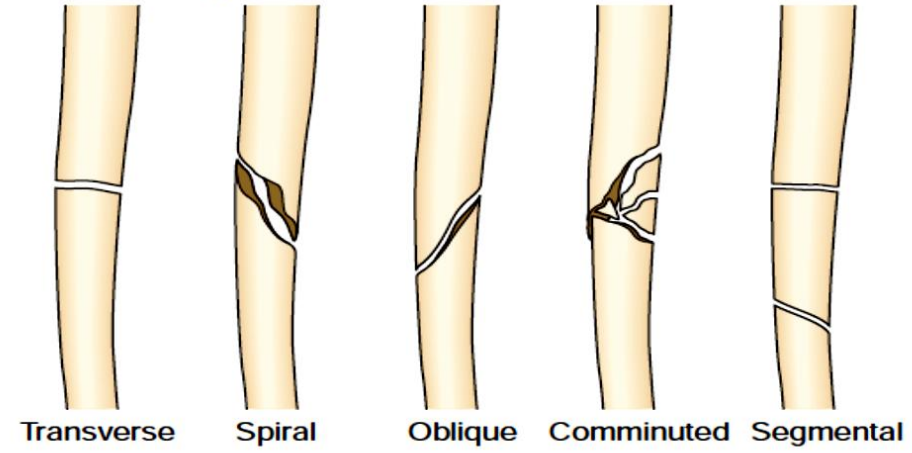
Fig.2 A transverse fracture of the femoral shaft over 50% displacement.



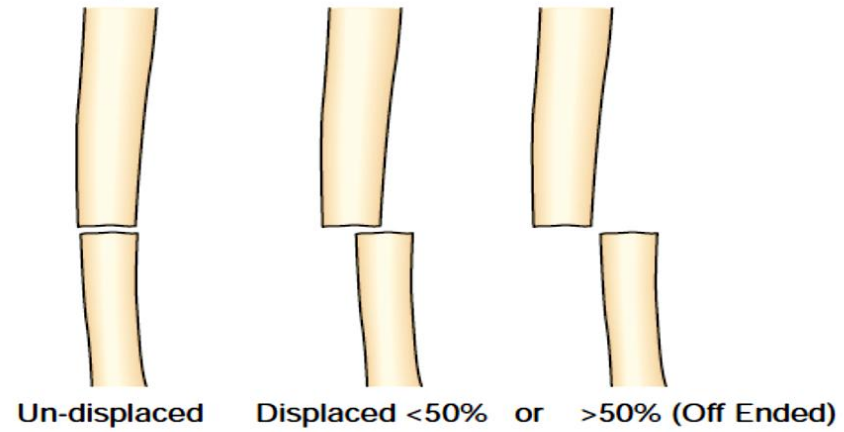
Fig.3 An intra-articular fracture of the lateral tibial plateau of the knee.



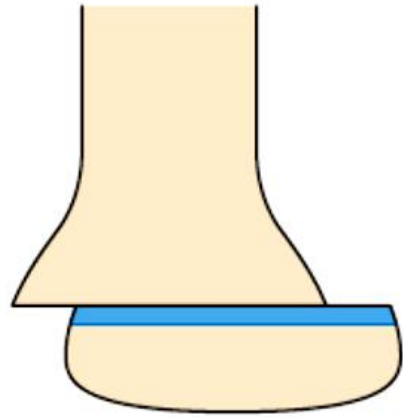
Common types of fracture



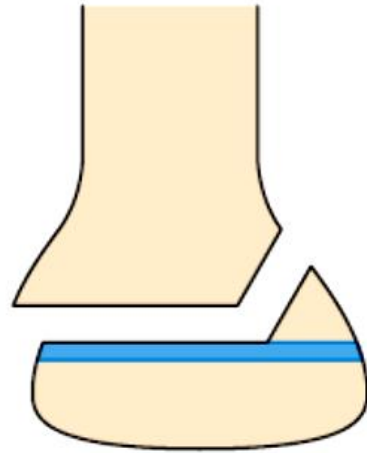
Displacement



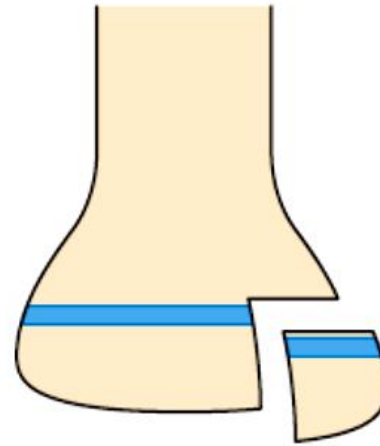
The location and description of many common fracture types.



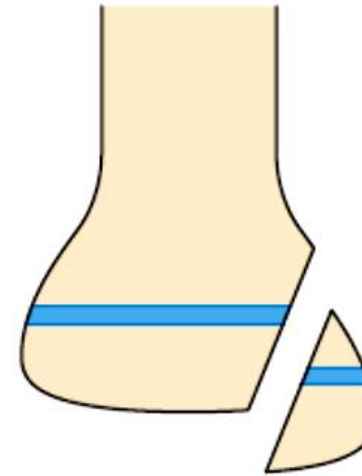
I
Epiphyseal
slip only



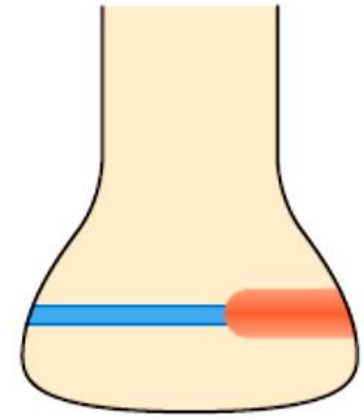
II
Fracture through
epiphyseal plate with
a fragment of metaphysis
attached



III
Fracture through
the epiphysis
extending into
epiphyseal plate



IV
Fracture of epiphysis
and metaphysis, crossing
the epiphyseal
plate



V
Damage to the
epiphyseal plate

The Salter-Harris classification of fractures.



Principles of Fracture Healing

- There are many differing ways of treating fractures.
 - However, they all have the same fundamental objectives: namely, the close approximation of uncontaminated, well-vascularized bone ends in a stable configuration that will maximize bone and soft tissue healing without deformity or loss of function.
 - In general, the fracture healing process takes approximately 8–10 weeks in adults but can take longer depending on the severity of the injury and whether it is an open fracture.
-

Compound Fractures

- These should be treated within 6 hours.
- However, new guidance from the combined orthopaedic and plastic surgery associations advises that these injuries can be left until the following morning (if admitted overnight and not grossly contaminated) when an experienced surgeon is available.

Definitive treatment should involve:

- Removal of all the damaged and dead tissue.
- Cleaning of the wound, with at least 3 litres of fluid depending on the degree of contamination ('the solution to pollution is dilution')
- Intravenous antibiotics
- Stabilization of the fracture to realign the bones to their anatomical position.
- Depending on the degree of contamination and soft tissue coverage, this may be by definitive or temporary fixation.
- Temporary external fixators are often attached by means of pins to the bones either side of the fracture site to allow access to the wound while imparting stability.

Intra-articular fractures

It is essential that the joint surface be reconstructed as accurately as possible to minimise the progression of secondary osteoarthritis. In certain fractures of the humeral and femoral heads, joint replacement may be the best option.

Conservative treatment

Closed fractures with good healing potential are usually treated with external stabilization using a simple bandage, plaster of Paris (POP) or a synthetic lightweight cast. In some cases where there is significant swelling the initial cast should not encircle the whole circumference of the limb.

Use of a 'back slab' of plaster allows the limb to swell and avoids the potential risk of a compartment syndrome. Once the acute swelling has settled, the cast can be completed to become a circumferential (full) cast. Displaced fractures often require manipulation under some form of anaesthetic prior to immobilization in POP.

Operative Treatment

- This involves the use of devices that are broadly divided into those that remain external to the skin and those that are internal (screws, plates and nails).



Fig. 27.25 A dynamic hip screw, a device used to secure hip fractures where the hip fracture is extracapsular as shown. The device is dynamic, as the screw can collapse down as the fracture heals.



Fig. 27.24 The use of an intramedullary nail to fix a tibial fracture

THANK YOU

